

MODULE 05

SQL DML - INSERT UPDATE DELETE

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL DML - INSERT UPDATE DELETE

DML Commands

```
INSERT: add new records
UPDATE: modify existing records
DELETE: remove records
SELECT: retrieve records
All DML operations can be rolled back.
```

INSERT

```
INSERT INTO students (first_name, gpa)
VALUES ('Arshith', 3.9);

-- Multiple rows
INSERT INTO students (first_name, gpa)
VALUES ('Rahul', 3.7), ('Priya', 3.8);
```

UPDATE

```
UPDATE students
SET gpa = 4.0
WHERE student_id = 1;

-- ALWAYS use WHERE clause!
-- Without WHERE, ALL rows are updated!
```

DELETE

```
DELETE FROM students WHERE student_id = 5;
DELETE FROM students WHERE gpa < 2.0;

-- Remove all rows (use TRUNCATE instead)
DELETE FROM students;
```

INSERT SELECT

```
-- Copy data between tables
INSERT INTO archive_students
SELECT * FROM students
WHERE enrollment_date < '2020-01-01';
```